

Asmita Pal

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Education

PhD in Computer Engineering

University of Wisconsin - Madison

Advisor : Prof. Joshua San Miguel

August'18 - Present

MS in Computer Engineering

Utah State University

Advisor : Prof. Koushik Chakraborty, Prof. Sanghamitra Roy

August'16 - August '18

B.Tech. in Electronics and Instrumentation Engineering

West Bengal University of Technology, India

July'10 - June'14

Publications

- Asmita Pal, Keerthana Desai, Rahul Chatterjee, Joshua San Miguel: **Camouflage: Utility-Aware Obfuscation for Accurate Simulation of Sensitive Program Traces**, published in *ACM Trans. Archit. Code Optim.* 21, 2, Article 36 (June 2024), 23 pages.
 - Mitali Soni, Asmita Pal, Joshua San Miguel: **As-Is Approximate Computing**, published in *ACM Trans. Archit. Code Optim.* 20, 1, Article 3 (March 2023), 26 pages.
 - Asmita Pal, Karthik Swaminathan, Subhankar Pal, Joshua San Miguel: **Characterizing Memory side channels in FHE applications** presented in *DISCC Workshop (held in conjunction with MICRO'22)*
 - Pramesh Pandey, Asmita Pal, Koushik Chakraborty, Sanghamitra Roy: **Reliability and Uniformity Enhancement in 8T-SRAM based PUFs operating at NTC**, published in *IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)* 2018.
 - Asmita Pal, Aatreyi Bal, Koushik Chakraborty, Sanghamitra Roy: **Split Latency Allocator: Process Variation-Aware Register Access Latency Boost in a Near-Threshold Graphics Processing Unit**, published in *J. Low Power Electronics* 13(3): 419-427 (2017).
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Awards

- Winner of Google's AI Sustainability Competition at **UW-Madison**
 - Sarah and David Epstein Teaching Fellowship [Awarded by Department of Electrical and Computer Engineering, **UW-Madison**]
 - Student Research Grant [Awarded by Office of Diversity, Inclusion and Funding at **UW-Madison**]
 - Anita Borg Grace Hopper Celebration 2019 Scholarship
 - Wisconsin Distinguished Fellow - Schneider [Awarded by Department of Electrical and Computer Engineering, **UW-Madison**]
 - Rajendra Prasad and Yasoda Mani Grandhi Fellow [Awarded by Department of Electrical and Computer Engineering, **Utah State University**]
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Experience

University of Wisconsin - Madison

WISCONSIN, UNITED STATES

Graduate Research Assistant

August '18 – Present

- Identify possible timing side-channel leakage in Network-on-Chips or interposer-based systems.
- Architectural characterization of memory side-channels in applications encrypted using Fully Homomorphic Encryption.
- Privacy preserving synthetic trace generation by leveraging program semantics.
- Identifying patterns in memory accesses and impact of learning techniques on the same.
- Investigating workloads amenable to approximate computing for approximation-aware architectures.

AMD Research

Research Intern

- Worked on LLC management guided by modelling data reuse behavior.

BOSTON, UNITED STATES

June 22 – Sept 22

AMD Research

Research Intern

- Worked on profiling memory bound, CPU-based AI applications to identify potential for acceleration.

BOSTON, UNITED STATES

September 20 – December 20

Utah State University

Graduate Research/Teaching Assistant

- Investigated power-performance trade-off characteristics for present day processors, including both GPU and CPU.
- Cross-layer analysis of the effects of process variation on a GPU register file, represented by 10T-SRAM cell, operating at Near Threshold Voltages.
- Studied efficacy of different thread criticality metrics in multi-core systems.
- Analyzing role of Deep Neural Machine learning in image recognition and speaker identification problems.
- **Teaching** - High Performance Computer Architecture(ECE-5750)

UTAH, UNITED STATES

August 16 – July 18

Infosys Technologies Limited

System Engineer

- Functioned as a team lead, and dealt with client deadlines and service delivery.
- Developed a project specific Automation Tool to enhance team productivity by amortizing the manual verification of device configurations.

PUNE, INDIA

July 14 – August 16

Recent Projects

Side-channel attacks for Mesh-based Network-on-Chips(NOC)

Dec 23 – Present

Supervisor : Prof. Joshua San Miguel

- Identifying contention in mesh interconnect.
- Applying temporal partitioning mechanisms to overcome observed contention.

Characterizing Memory Side Channels of FHE Applications

Dec 21 – Dec 23

Supervisor : Prof. Joshua San Miguel

- Analyzing side channels in applications encrypted using FHE (Fully Homomorphic Encryption).
- Studying memory access patterns to understand vulnerability of operations in FHE domain.

Privacy-preserving Traces by leveraging Program Semantics

Jan 20 – Oct 23

Supervisor : Prof. Joshua San Miguel

- Characterized inherent architectural properties of program structures that influence program behavior, captured in traces.
- Proposed a trace-based obfuscation mechanism focused on program-semantic specific obfuscation techniques, with a security-conservative mindset.
- Developed a generalized threat model based on input indistinguishability to ensure input secrecy.

Analyzing patterns in memory accesses

Sep 18 – Dec 19

Supervisor : Prof. Joshua San Miguel

- Understanding impact of program structure on memory accesses and leveraging that to generate synthetic traces.
- Applying sequence prediction problem of natural language processing framework to memory access sequences.

Architectural Support for Anytime Approximate Computing

Jan 19 – Nov 19

Supervisor : Prof. Joshua San Miguel

- Design of speculation based architecture for anytime approximate computing.
- Characterizing approximate applications for an interruptive system to accommodate different levels of accuracy with runtime.

Digit Recognition on FPGA using Multilayer Perceptron

Sep '19 – Dec '19

Supervisor : Prof. Parmesh Ramanathan

- Design of processor, memory and accelerator modules to enable storage of a pre-trained model and offload computations to accelerator.
- Facilitating timely image capture and display of recognized image on the 7-segment display.

Precise encoding in Stochastic Computing for Neural Networks

Sep '18 – Dec '18

Supervisor : Prof. Younghyun Kim

- Proposed of a new encoding scheme for Stochastic Numbers and implemented this new encoding format in MAC module of a small scale Neural Network.
- Analyzed the energy-accuracy trade-off for LeNet300-100 and observed upto 5X energy degradation.

Design of a Process-Variation Aware Low Power GPU

Jan '17 – July '17

Supervisor : Prof. Koushik Chakraborty, Prof. Sanghamitra Roy

- Explored the factors affecting the operation of a GPU at Near Threshold Voltages, especially the role of process variation on register files.
- Analyzed the performance variation in GPGPU applications from AMD APP SDK suite, with varying register latency.
- Proposed a low overhead technique to overcome the effects of process variation on register latency. Exhibited ~36% reduction in GPU energy consumption with marginal area and power overheads.

Speaker Diarization using Deep Neural Networks

Sep '17 – Dec '17

Supervisor : Prof. Todd Moon

- Extracted features from a conversation and use Deep Neural Networks to separate this audio signal into its homogeneous components according to speaker identity.
- Presented a comparative performance evaluation in the speaker diarization framework, using deep neural networks and convolutional neural networks (max prediction accuracy ~89%).

Critical Thread Predictor

Jan '17 – April '17

Supervisor : Prof. Koushik Chakraborty

- Illustrated the statistical coherence of thread criticality metrics based on cache-misses and synchronization bottlenecks and discussed the scope of performance optimization.
- The simulations were executed on a multi-core CPU, modelled on Intel Nehalem architecture, running applications from Splash2 Benchmark suite.

GPU Power and Area Analysis

Jan '17 – April '17

Supervisor : Prof. Sanghamitra Roy

- Synthesis, place and route for basic pipe stages of a GPU, to study of change in power and area and timing violation, with frequency and technology node variation.

Design of a Global Router

Aug '16 – Dec'16

Supervisor : Prof. Sanghamitra Roy

- Developed a VLSI floorplan based on simulated annealing algorithm.
- Implemented a global routing engine using A* search algorithm

Professional Service**Invited Talks**

- University of Central Florida - Computer Architecture Coordination Seminar Series, November 2023

Artifact Evaluation Committee

- IEEE/ACM International Symposium on Microarchitecture 2023 ([MICRO 2023](#), 2022)
- Architectural Support for Programming Languages and Operating Systems ([ASPLOS 2022](#))
- Conference on Machine Learning and Systems ([MLSys 2020](#))
- Architectural Support for Programming Languages and Operating Systems ([ASPLOS 2020](#))

Teaching

- TA for ECE352(Digital System Fundamentals) and ECE757(Advanced Computer Architecture)
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Research Interests

- Computer Architecture
 - Synthetic Trace Generation
 - Data Privacy
 - Caches and Memory Access Patterns
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Relevant Coursework

University of Wisconsin - Madison

WISCONSIN, UNITED STATES

Advanced Computer Architecture, Introduction to OS, Embedded Computing Systems, Security and Privacy

Utah State University

UTAH, UNITED STATES

High Performance Computer Architecture, Parallel Computer Architecture, VLSI Design Automation, Neural Networks, Advanced Algorithms